GigaWhat Energy

GigaWhat Electricity
Controlled procedure

PR-ELEC-070 Live Working on Low Voltage Equipment

DRAFT: NOT APPROVED FOR USE.

Field	Value
Document ID	PR-ELEC-070
Version	1 (draft)
Effective from	Not yet approved
Review due	Not yet approved
Owner	Electricity Safety Manager
Approved by	Not yet approved
Applies to	All GigaWhat Electricity sites: low voltage equipment at distribution transformers and ring main units

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out when live working on low voltage (LV) equipment may be justified and how it must be done, so that the risk of electric shock and arc flash burns is controlled.

> NOTE: This is a first draft for consultation. Items marked "Drafting note" are still to be agreed.

2 Scope

This procedure applies to LV (400/230 V) equipment supplied from distribution transformers, including the LV boards fed by TX-BRK-01 at ESS-BRK and TX-ORC-01 at ESS-ORC, and LV equipment in substations housing ring main units. Live working on HV equipment is prohibited under PR-ELEC-001 and is not covered.

3 References

- PR-ELEC-001 Distribution Safety Rules - Summary for Operational Staff
- PR-ELEC-012 Isolation and Earthing of 11 kV Circuits
- PR-ELEC-020 Permit-to-Work and Sanction-for-Test (Electrical)
- PR-ELEC-063 Substation Access, Keys and Visitors

- PR-CORP-001 Reporting Incidents and Near Misses
- PR-CORP-004 Lone Working
- HSE HSG85 Electricity at work: Safe working practices
- HSE HSR25 Memorandum of guidance on the Electricity at Work Regulations 1989
- Electricity at Work Regulations 1989
- Management of Health and Safety at Work Regulations 1999

4 Definitions

Live working: work on or near uninsulated LV conductors that remain live, where there is a risk of contact or of a short circuit.

Live Working Authorisation (LWA): a written authorisation issued by an Authorised Person (HV) for one specific live working task.

LV live working endorsement: an endorsement on a Competent Person's authorisation, granted by the Electricity Safety Manager after training and assessment.

Accompanying person: a second Competent Person (HV), trained in LV rescue, who watches the work and does not take part in it.

Prospective fault current: the current that would flow in a short circuit at the point of work.

5 Responsibilities

Electricity Safety Manager: grants LV live working endorsements and approves the tools and personal protective equipment used.

Senior Authorised Person (HV): approves live working where the prospective fault current exceeds 20 kA.

Authorised Person (HV): decides whether live working is justified and issues the Live Working Authorisation.

Competent Person (HV) with LV live working endorsement: carries out the work.

Accompanying person: watches the work continuously, stops it if needed and carries out rescue.

Control Engineer: is informed before work starts and when it finishes.

6 Procedure

6.1 Deciding whether live working is justified

> NOTE: Live working is a last resort. The Electricity at Work Regulations 1989 allow work on or near live conductors only where it is unreasonable for them to be dead, reasonable for the work to be done live and suitable precautions are taken.

6.1.1 First plan to make the equipment dead and work under PR-ELEC-020. Record why that is not reasonable before considering live working.

6.1.2 Consider live working only for connecting or disconnecting LV cables at a live LV board, replacing LV fuses, connecting temporary generators and taking measurements.

6.1.3 Do not work live in rain or wet conditions, in a confined space, where insulation is damaged or where live parts cannot be shrouded.

6.1.4 Drafting note: the list of permitted tasks in 6.1.2 is to be confirmed with the Senior Authorised Person (HV).

6.2 Authorisation and planning

6.2.1 As Authorised Person (HV), complete a task risk assessment and issue a Live Working Authorisation stating the justification, method, shrouding and personal protective equipment.

6.2.2 Find the prospective fault current at the point of work. Where it exceeds 20 kA, as on the LV board fed by TX-BRK-01, obtain Senior Authorised Person (HV) approval.

6.2.3 Assign an accompanying person. Lone live working is prohibited.

6.2.4 Inform the Control Engineer before work starts.

> HOLD POINT: Work must not start until the Authorised Person (HV) has issued the Live Working Authorisation and briefed both the Competent Person (HV) and the accompanying person on site.

6.2.5 Drafting note: the arc rating required at ESS-BRK is to be confirmed by an arc flash study.

6.3 Personal protective equipment and tools

6.3.1 Wear arc-rated clothing of at least 12 cal/cm², an arc-rated face shield and balaclava. Drafting note: 25 cal/cm² proposed for ESS-BRK, subject to 6.2.5.

6.3.2 Wear class 0 insulating gloves with leather over-gloves, tested within the last 6 months, and air test them before use.

6.3.3 Use only insulated tools approved by the Electricity Safety Manager and inspect them before use.

6.3.4 Use an approved voltage indicator and proving unit, proving the indicator before and after each test.

6.3.5 Have an insulated rescue hook, first aid kit and defibrillator at the work area.

6.4 Preparing the work area

6.4.1 Set up barriers to keep everyone except the working pair at least 2 m from the work.

6.4.2 Lay an insulating mat in front of the equipment.

6.4.3 Remove covers only where needed, and shroud every live part that is not being worked on.

6.4.4 Identify the circuit by its label and by test before starting.

6.4.5 Position the accompanying person within reach of the means of isolation, and confirm they know how to operate it.

6.5 Carrying out the work

> WARNING: A tool or conductor bridging two phases, or a phase and earth, on an LV board fed by a large distribution transformer can cause an arc flash that results in severe burns, blindness or death.

6.5.1 Work on one phase at a time, insulating each conductor end before moving on.

6.5.2 Connect the neutral first and disconnect it last.

6.5.3 Keep your body out of line with the conductors being worked on as far as you can.

6.5.4 As accompanying person, watch continuously and stop the work if anything changes.

6.5.5 Stop work at once if conditions change, insulation is damaged or anyone enters the barriers.

6.5.6 When the work is complete, remove the shrouds, refit the covers and inform the Control Engineer.

6.6 Emergency

6.6.1 If someone receives an electric shock, isolate the supply or pull them clear with the insulated rescue hook. Do not touch them directly.

6.6.2 Call 999 for any injury or fire.

6.6.3 Cool burns with clean running water for at least 20 minutes.

6.6.4 Inform the Control Engineer and report the event under PR-CORP-001.

6.6.5 Ask members of the public who report a power cut or damaged power lines to call 105.

7 Records

- Live Working Authorisations and risk assessments, kept for six years.
- LV live working endorsements, kept by the Electricity Safety Manager.
- Insulating glove test records.

8 Revision history

Version	Date	Change
1 (draft)	Not yet approved	First draft for consultation.